Spectral-Adapted Vision Transformer for Multi-Class Crop Disease Classification from Sentinel-2 Multispectral Imagery

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Abstract

Random image-level splitting can lead to optimistic performance estimates in satellite image classification because spatially neighbouring patches are not statistically independent. This study examines the magnitude of this effect in multispectral crop disease classification and reassesses a spectral-adapted Vision Transformer under a leakage-controlled evaluation protocol. Using the ICPR 2024 *Beyond Visible Spectrum* Sentinel-2 benchmark, spatially dependent and duplicate patches are grouped into indivisible units to construct a group-disjoint train–validation–test partition. A conventional stratified random split is retained as a controlled reference using the same cohort, training code, random seeds, epoch budget, learning rates, and augmentation settings. Across three seeds, random splitting increases macro-F1 by **0.138**–**0.273**, corresponding to a relative overestimation of 25–53%. A validation-only learning-rate audit changes the selected rate in five of six architecture–strategy configurations and improves an under-trained EfficientNet-B3 baseline from **0.213** to **0.472** macro-F1. Under leakage control, the highest mean macro-F1 is $\mathbf{0.600 \pm 0.088}$, with overlapping confidence intervals across the evaluated architectures. Train-only per-band normalisation consistently outperforms ImageNet-replicated normalisation, whereas mixed-sample augmentation provides no measurable macro-F1 improvement. A pre-specified four-band configuration also performs within one standard deviation of the corresponding six-band configuration. These findings show that spatial partitioning and architecture-specific optimisation materially influence apparent model performance and should be controlled before drawing conclusions about architectural or training-strategy superiority.

Keywords: Crop disease classification, Data leakage, Evaluation protocol, Multispectral imagery, Sentinel-2, Spatial cross-validation, Vision Transformer

1 Introduction

Crop diseases and pests account for an estimated 20–40% of annual global crop yield losses, at an economic cost of roughly USD 220 billion per year [1]. Early and spatially explicit detection is therefore a persistent priority for agricultural monitoring. Field inspection by trained agronomists does not scale to regional coverage [2], and satellite remote sensing has become the practical alternative [3]. The Sentinel-2 constellation delivers 13 spectral bands from the visible to the shortwave infrared at 10 m to 60 m ground sampling distance, with a five-day revisit interval and open access [4]; several of these bands respond to the physiological changes that disease induces in vegetation [5].

Deep learning has been applied extensively in this setting [6]. Vision Transformers (ViTs) [7] have been adopted because global self-attention relates distant regions of a scene, which convolutional kernels model only indirectly [8, 9]. Extending an RGB-pretrained ViT to more than three spectral channels is technically simple, and models built in this way report strong results on public benchmarks.

Those results, however, are almost always obtained under an evaluation protocol that is not appropriate for geospatial data. Stratified random splitting at the level of individual image patches assumes that patches are independent draws, an assumption that satellite imagery does not satisfy. Neighbouring patches share illumination, atmospheric state, phenological stage, soil background and field boundaries, and they frequently cover overlapping ground. A random split consequently places statistically dependent samples on both sides of the train/test boundary. Outside the crop-disease literature this consequence is established: spatially blind validation overestimates predictive skill in ecological mapping [10] and in remote-sensing classification [11], and leakage of this kind has been identified as a leading cause of irreproducible results across machine-learning-based science [12]. Group-aware and block-based cross-validation are the accepted remedy [13]. For satellite-based crop disease classification, however, the magnitude of the distortion has not been measured under controlled conditions. The present study provides that measurement and then reports what a spectral-adapted ViT achieves once the distortion is removed. The starting point is a conventional pipeline: a ViT-B/16 whose patch-embedding projection is extended from three to six Sentinel-2 channels, combined with an imbalance-oriented training strategy. This pipeline is not offered as a new state of the art but treated as a representative object of study, evaluated under a protocol designed so that its reported performance cannot be inflated by partition artefacts.

Two methodological problems are addressed. The first is spatial leakage. The benchmark is audited at the level of georeferenced patch geometry, and every pair of patches that overlaps, lies within 500 m of another, falls in the same 10 km block or shares a content hash is merged into a single indivisible group, from which a group-disjoint partition is constructed. A *reference arm* then reproduces the conventional random image-level split on the same cohort, code, seeds, epoch budget, learning rates and augmentation, so that the partition rule is the only variable differing between the two arms. The resulting paired design yields an estimate of the leakage effect with a confidence interval rather than an assertion that the effect exists.

The second problem arises in architecture comparison. When several architectures are compared under one shared learning rate, the comparison measures compatibility with that rate as much as architectural merit. An earlier configuration of this study illustrates the point: a rate suited to LayerNorm-based transformers left a BatchNorm-based EfficientNet-B3 severely under-trained at 0.213 macro-F1, a value that the loss curves identify as unconverged rather than merely low. A pre-specified learning-rate search, applied identically to every architecture×strategy cell and scored on validation data only, changes the selected rate in five of six cells and raises that configuration to 0.472. Omitting this step risks attributing to architecture what belongs to optimiser settings, as has been shown for ResNets under modernised training recipes [14].

Three contributions follow. The first is a pair of quantified methodological effects. Under a design in which only the partition rule varies, random image-level splitting inflates macro-F1 by 0.138 to 0.273 (25–53% relative) on this benchmark, with all three tested configurations nominally significant at $p < 0.05$; Section 6.1 reports the Holm-adjusted values. To our knowledge this is the first controlled measurement of the effect for satellite-based crop disease classification. The accompanying learning-rate audit uses a pre-specified grid applied identically to all six architecture×strategy cells, and the complete search table is reported, including candidates that were rejected. The second contribution revises three design choices on evidence rather than on argument: a 2×2 ablation shows that train-only per-band normalisation gives higher mean macro-F1 than ImageNet-replicated statistics for the NIR and SWIR channels, mixed-sample augmentation provides no measurable macro-F1 benefit once leakage is controlled, and a pre-declared four-band configuration (B2, B3, B4, B8) matches the six-band configuration within one standard deviation. The last of these has practical value for data acquisition and storage, although an undetected difference is not evidence of equivalence. The third contribution concerns the limits of the benchmark itself: leave-one-region-out and leave-one-season-out evaluation are shown to be infeasible from the available metadata, and are reported as such rather than approximated.

The remainder of the paper is organised as follows. Section 2 reviews related work, including the validity literature that motivates the protocol. Section 3 describes the dataset and the leakage audit. Section 4 presents the model and training strategy. Section 5 specifies the evaluation protocol, which is deliberately separated from the method. Section 6 reports results, Section 7 discusses them, Section 8 states limitations, and Section 9 concludes.

2 Related Work

2.1 Evaluation validity and leakage in geospatial machine learning

The statistical difficulty with random splitting of spatial data is autocorrelation: nearby observations resemble one another more than distant ones, so a randomly held-out sample is frequently a near-replicate of a training sample. Roberts et al. [13] formalise this for temporally, spatially and hierarchically structured data and recommend blocking strategies that respect the dependence structure. The empirical consequences have been documented across several domains. Ploton et al. [10]

show that large-scale ecological mapping models with strong random-split statistics lose most of their apparent skill under spatial validation, and Karasiak et al. [11] demonstrate the same bias for remote-sensing image classification. Kattenborn et al. [15] report that spatially autocorrelated training and validation samples inflate the assessed performance of convolutional networks by up to 28% relative to spatial cross-validation, which establishes that the problem is not confined to the shallow learners studied earlier. Kapoor and Narayanan [12] document leakage across 17 scientific fields and 294 affected papers, and identify train-test dependence of this kind as a recurring failure mode.

Remedies are available. Spatial and environmental blocking schemes are implemented in standard tooling [16], and Meyer and Pebesma [17] address the complementary question of where a fitted model may legitimately be applied, subsequently showing that widely circulated global ecological maps are frequently evaluated outside their area of applicability [18]. Schratz et al. [19] establish a point that bears directly on the protocol adopted here: hyperparameter tuning and performance assessment interact with the resampling scheme, so tuning decisions taken under a spatially blind scheme can themselves bias a comparison. This is the reason model selection in the present study is conducted under the same protocol used for reporting, and the reason the complete search is published rather than only its outcome.

A benchmark reporting random-split figures therefore carries an unquantified optimistic bias by default. The magnitude of that bias, however, is dataset-specific and cannot be inferred from other domains; it must be measured on the benchmark in question, under a design in which the partition rule is the only quantity that changes. No such measurement has been reported for satellite-based crop disease classification. Section 6.1 provides it.

2.2 Remote sensing for crop disease detection

Multispectral and hyperspectral sensors capture reflectance across wavelength bands that correlate with disease-induced physiological change, and vegetation indices derived from red, red-edge and near-infrared reflectance, beginning with NDVI [20], remain the standard descriptors of canopy condition. Sentinel-2 combines sufficient spatial resolution, a short revisit interval and open access [4], and has become the default platform for operational agricultural monitoring. Index-based approaches remain competitive at this resolution: Zheng et al. [21] derive a Sentinel-2 spectral index that separates yellow-rust severity levels in winter wheat, and comparable indices underpin much of the operational literature. Learned approaches extend this line. Shi et al. [5] couple Sentinel-2 time series with convolutional models to separate yellow rust from nitrogen deficiency in wheat, which illustrates both the value of the data and the difficulty of distinguishing stressors whose spectral signatures overlap. Where spatial or spectral resolution limits discrimination, generative super-resolution has been explored as a preprocessing route [22].

2.3 Vision Transformers for multispectral data

Convolutional networks defined the previous generation of remote-sensing classifiers. Deep plain stacks [23] and residual networks [24] were followed by densely connected [25], mobile-oriented [26], compound-scaled [27, 28] and design-space-searched [29] variants, with ConvNeXt [30] subsequently modernising the family.

The Vision Transformer [7] replaces convolutional inductive bias with global self-attention [31]. Data-efficient training [32] and masked-image pretraining [33] reduced its dependence on very large labelled corpora, and hierarchical variants such as Swin [34] restore locality through shifted windows at reduced cost. Bazi et al. [35] were among the first to show that these models transfer to remote-sensing scene classification without convolutional stems. Within remote sensing, plain ViTs have been scaled into domain foundation models [8], and SatMAE [9] demonstrates that masked pretraining on temporal and multispectral satellite imagery yields representations better matched to the modality than ImageNet initialisation. SpectralFormer [36] models inter-band dependencies explicitly for hyperspectral classification. For agricultural disease monitoring, Das et al. apply a transformer-embedded attentive CNN to spectral analysis of rice blast [37] and a multigranular Swin architecture to hierarchical mixing of spectral indices for the same disease [38]. These studies were evaluated on different data under different protocols, so their numerical results are not comparable with those reported below; they are cited for the modelling approaches they establish.

2.4 Class imbalance and mixed-sample augmentation

Class imbalance biases decision boundaries toward majority classes [39]. Loss-level remedies include Focal Loss [40], class-balanced reweighting by effective sample number [41] and label-distribution-aware margins [42]; label smoothing [43] additionally regularises overconfident targets. At the data level, MixUp [44] and CutMix [45] synthesise convex or spatially spliced combinations of training pairs, and are widely reported to improve generalisation [46].

Whether such gains transfer across architectures is less well established. Rana et al. [47] find that strong domain transformation of vegetation imagery affects detectors unevenly, improving one configuration while degrading another, with performance recovering only when original and transformed data are combined. Their task and transformation differ from those examined here, so the finding is not evidence for a general rule; it does indicate that the benefit of mixed-sample augmentation is a property of a particular configuration rather than of the technique, which is the question Section 6.5 addresses under leakage control.

3 Dataset and Leakage Audit

The dataset and its audit precede the method, because the audit determines the evaluation protocol under which the method is assessed.

Table 1 Selected Sentinel-2 bands. Protocol A uses all six; Protocol B uses the four bands available at 10 m native resolution in every sample folder.

Band	Centre (nm)	GSD (m)	Physical relevance	Protocol B
B2 (Blue)	490	10	Pigment and aerosol response	✓
B3 (Green)	560	10	Chlorophyll reflectance peak	✓
B4 (Red)	665	10	Chlorophyll absorption, stress	✓
B5 (Red edge)	705	20	Fine-scale stress transition	—
B8 (NIR)	842	10	Canopy structure, leaf area index	✓
B11 (SWIR)	1610	20	Canopy and soil moisture	—

3.1 Data source and classes

Task 2 of the ICPR 2024 *Beyond Visible Spectrum: AI for Agriculture* competition [48] provides Sentinel-2 surface-reflectance patches over major crop production zones in China, acquired across four seasonal windows. Each patch covers approximately 2800 m × 2800 m at 264 × 264 pixels and 10 m ground sampling distance. Four disease categories are labelled: Aphid, Blast (*Magnaporthe* spp.), Rice Planthopper (RPH) and Rust (*Puccinia* spp.). The class distribution is markedly imbalanced, with 495 RPH, 290 Aphid, 75 Blast and 40 Rust samples, a ratio of 12.4:1 between the largest and smallest class.

3.2 Integrity verification

The raw archive was re-inventoried before any modelling, because an earlier pass had recognised only 879 of the 900 sample folders as six-band complete. The shortfall proved to be an artefact of file handling rather than of the dataset: band-name parsing had to be made case-insensitive, to accept `.tif`, `.tiff` and `.jp2` extensions, to search nested directories, and to distinguish B8 (842 nm, 10 m) from B8A (865 nm, 20 m). After correction, all 900 folders were confirmed complete for the six selected bands, and the class counts match the competition specification exactly. Band B10 is absent from all 900 folders, as expected for Level-2A products, and is excluded.

Two protocols are declared before training and used throughout. Protocol A uses B2, B3, B4, B5, B8 and B11 on all 900 patches. Protocol B uses the four bands present at 10 m native resolution in every folder, namely B2, B3, B4 and B8, on the same 900 patches. Protocol B is defined by band availability and a pre-specified agronomic ordering, never by observed predictive performance. Because both protocols resolve to the identical 900-patch cohort, they share one partition, which isolates the effect of band removal from any effect of cohort restriction.

3.3 Grouping rule

For every patch the audit reads the coordinate reference system, affine transform, projected bounds, patch centroid, any available acquisition tags and a content hash of the selected bands. Two patches are assigned to the same *primary group* if their projected footprints intersect, their centroids lie within 500 m of each other, they fall in the same 10 km spatial block, or their selected-band content hashes are identical.

Table 2 Composition of the two partitions. Both use the identical 900-patch cohort and identical test size. The grouped partition is the primary protocol; the random partition is the reference arm used to measure leakage.

Subset	Aphid	Blast	RPH	Rust	Total
Group-disjoint partition (97 groups, 0 crossing)					
Train	196	67	352	26	641
Validation	29	0	45	5	79
Test	65	8	98	9	180
Random image-level partition (84 of 97 groups crossing)					
Train	203	52	346	28	629
Validation	29	8	50	4	91
Test	58	15	99	8	180

Groups are formed as connected components under these relations using a disjoint-set structure, so grouping is transitive: if a overlaps b and b lies within 500 m of c, all three are indivisible. All 900 patches carry valid projected bounds. No exact content-duplicate cluster was found, so the content-hash criterion, although declared in advance, never contributes to a merge; the dependence in this benchmark is geometric rather than the result of literal file duplication. The procedure yields 97 primary groups across 900 patches, a mean of 9.3 patches per group, which indicates how strongly the benchmark is spatially clustered.

Group support is highly uneven across classes: Aphid occupies 43 independent groups, RPH 44, Rust 10, and Blast only 2. This is a property of the data rather than of the audit procedure, and it constrains both the partition and what any model can be expected to learn about Blast (Section 5).

3.4 Group-disjoint partition

The partition targets 70/10/20 while enforcing three constraints: groups are never split across subsets; all four classes must be present in train and test; and a class may be absent from validation only if it occupies fewer than three independent groups. An initial candidate was rejected because its test subset contained no Blast samples. Since Blast occupies exactly two groups, group-disjoint Blast support in train, validation *and* test is mathematically impossible, so the final partition places Blast in train and test and accepts its absence from validation. We disclose this rather than relaxing group disjointness.

The resulting partition (Table 2) has no group crossing any subset boundary and is fixed by the hash `816a211e...`, which every subsequent experiment verifies before training, so that no run can execute on a different partition without detection.

3.5 Reference arm: the conventional random partition

To measure the effect of the partition rule, the conventional protocol is reproduced on the identical cohort: stratified random splitting at the level of individual patches, at the same 70/10/20 ratio. This arm is a controlled comparator, not a competing method. Everything except the partition rule is held fixed, including the code paths,

Table 3 Train-only per-band reflectance statistics, computed independently for each arm. The close agreement confirms that the two arms differ in partition rule rather than in input scaling.

	Grouped arm ($n = 641$)		Random arm ($n = 629$)	
Band	Mean	SD	Mean	SD
B2	0.2348	0.2469	0.2301	0.2415
B3	0.2554	0.2298	0.2504	0.2247
B4	0.2598	0.2238	0.2551	0.2195
B5	0.3029	0.2193	0.2978	0.2154
B8	0.3778	0.1933	0.3738	0.1901
B11	0.3383	0.1573	0.3373	0.1571

the three seeds, the 60-epoch budget, the learning rates selected in Section 5.2 and the augmentation.

Two safeguards make the comparison defensible. Normalisation statistics are recomputed from the training subset of each arm separately, so the random arm receives no normalisation advantage; the resulting per-band means differ from those of the grouped arm only in the third decimal place (Table 3), which confirms that the two arms are not being compared across different input scalings. The pipeline also raises an error if the random partition fails to produce group crossing, since an arm without crossing would not be a leakage comparator at all. In practice the random partition places 84 of the 97 groups on both sides of the train/test boundary, and it is this crossing that Section 6.1 quantifies.

3.6 What the metadata does and does not support

Region-held-out and season-held-out evaluation are the natural next tests for geospatial work. Both were attempted on this cohort, and neither is possible; the outcome is reported here rather than approximated.

Leave-one-season-out. No acquisition date is recoverable for any of the 900 patches. No temporal-generalisation claim is therefore made anywhere in this study, and season was not inferred from class or location.

Leave-one-region-out. No official tile or administrative region identifier accompanies the data, so four candidate macroregions were derived by label-blind k-means clustering of patch centroids, giving groups of 570, 299, 23 and 8 patches. A prespecified eligibility test requiring the held-out partition to contain all four classes rejected every one of the four candidate folds: the held-out regions contain, respectively, Blast and RPH only; Aphid and Rust only; Aphid only; and Rust only. No valid fold exists. Disease class and geography are strongly confounded in this benchmark, so a coordinate-derived regional holdout would measure class absence rather than geographic transfer.

This is a substantive property of the benchmark rather than a gap in the experiments, and Section 8 states it as such. Substituting a weaker proxy would produce a figure resembling a generalisation result without being one.

4 Method

The model follows a standard and deliberately unexceptional recipe; the contribution of this study lies in the protocol under which it is evaluated, not in architectural novelty. Figure 1 summarises the framework. Six normalised bands form a $6 \times 224 \times 224$ input tensor. The patch-embedding projection, extended from three to six channels, partitions the image into non-overlapping 16×16 patches and maps each to a 768-dimensional token, giving 196 tokens to which a class token and learnable position embeddings are prepended. Twelve encoder blocks with twelve attention heads and feed-forward hidden dimension 3,072 process the resulting sequence, and a single linear layer maps the final class-token representation to four logits.

Two aspects of the figure require comment, since both are decisions the study revises rather than assumes. The patch-embedding projection is drawn as an experimental variable rather than as a fixed component: the literature freezes this layer on the argument that replicated ImageNet filters should be preserved, but that argument has not been tested, and Section 6.4 reports the ablation that examines both treatments. Normalisation statistics are likewise computed from the training subset alone rather than inherited from ImageNet, a choice adopted on the evidence of the same ablation and not on prior practice. Augmentation is applied along the training path only, so that the validation and test paths are deterministic and no stochastic component can enter the reported measurements.

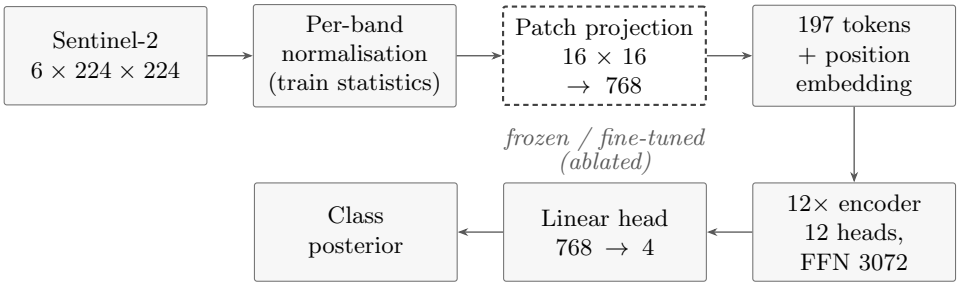


Fig. 1 The spectral-adapted ViT-B/16 framework under Protocol A.

4.1 Preprocessing and normalisation

Reflectance is scaled to $[0, 1]$ by clipping and dividing by the standard Sentinel-2 factor,

$$X_{\text{norm}} = \text{clip}\left(\frac{X_{\text{raw}}}{10000}, 0, 1\right), \tag{1}$$

after which the selected bands are stacked into a $224 \times 224 \times C$ tensor ($C \in \{4, 6\}$) by bilinear resampling from 264×264. Bands B5 and B11 are supplied by the dataset provider pre-resampled to the 10 m grid.

Each channel is then standardised with statistics computed from the training subset only:

$$X_{\text{std}}^{(b)} = \frac{X_{\text{norm}}^{(b)} - \mu_b}{\sigma_b}, \tag{2}$$

where μ_b and σ_b are the per-band mean and standard deviation in Table 3. This differs from the ImageNet-replicated normalisation used in much prior work, in which a single RGB-derived mean and standard deviation are applied to every channel including NIR and SWIR. That choice has been asserted rather than tested, so it is treated here as an experimental question; Section 6.4 reports the comparison, and the per-band variant is adopted on the strength of that evidence.

4.2 Spectral adaptation of the patch embedding

The backbone is ViT-B/16 [7] pretrained on ImageNet [49]. The patch-embedding convolution is extended from $\mathbb{R}^{768\times3\times16\times16}$ to $\mathbb{R}^{768\times C\times16\times16}$ by the replication-and-rescaling initialisation provided by `timm` [50], which preserves the pretrained response magnitude while accommodating additional channels. Whether this layer should subsequently be frozen or fine-tuned is likewise an empirical question. Section 6.4 reports a 2×2 ablation over the two treatments and the two normalisation schemes; the normalisation choice is resolved by that ablation, whereas the freezing choice is not separated at three seeds and is reported as unresolved.

4.3 Training strategy

Two training strategies are compared throughout, and both are applied identically to every architecture.

CE. Standard cross-entropy, with no mixed-sample augmentation and no weighted sampling.

WLS+MSDA. An imbalance-oriented combination of three components. The first is weighted random sampling with replacement, with weight $w_i \propto 1/N_{c(i)}$ normalised to unit mean. The second is a weighted label-smoothing cross-entropy,

$$\mathcal{L}_{\text{WLS}} = -w_c \sum_{k=1}^{C} q_k \log p_k, \qquad q_k = \begin{cases} 1-\varepsilon & k = c \\ \varepsilon/(C-1) & k \neq c \end{cases} \tag{3}$$

with $\varepsilon = 0.1$ and class weights $w_c \propto 1/N_c$ normalised to unit mean [43]. The third is batch-level MixUp [44] ($\alpha = 0.4$) and CutMix [45] ($\alpha = 0.7$), applied with probability 0.5 and switching between the two with equal probability; when mixing is active, the soft mixed target replaces q_k.

Spatial augmentation comprises horizontal and vertical flips, 90° rotations, shift-scale-rotate and brightness/contrast jitter [51]. Every transformation is applied identically to all bands of a sample rather than independently per channel, since per-channel jitter would perturb the ratios between bands on which vegetation indices and, by extension, the spectral signature of a diseased canopy depend. The geometric transformations are appropriate for this modality because a satellite patch has no canonical orientation: the same field observed on a different overpass may appear

rotated or reflected with no change in its physical content. Augmentation is confined to the training path, so that validation and test are evaluated on unmodified inputs and no stochastic component enters the reported measurements. Optimisation uses AdamW [52] with weight decay 0.05 and cosine annealing, with the learning rate selected per architecture and strategy by the procedure of Section 5.2 rather than shared across configurations. Algorithm 1 states the complete procedure.

Algorithm 1 Leakage-controlled multispectral training and evaluation

1: Inventory sample folders; retain samples containing every band of the declared protocol.
2: Read CRS, affine transform, projected bounds, centroid, acquisition tags and selected-band content hashes.
3: Merge overlapping, proximate ($\leq$ 500 m), same-block (10 km) and content-duplicate samples into primary groups.
4: Build group-disjoint train/validation/test subsets; require all classes in train and test; permit validation to omit a class only if it occupies fewer than three groups.
5: Compute per-band reflectance mean and standard deviation from the training subset only.
6: **for** each declared seed $s \in \{42, 123, 2024\}$ **do**
7: Initialise the ImageNet-pretrained backbone with `in_chans`$= C$ and a four-class head.
8: Apply spatial augmentation to training samples only; for WLS+MSDA additionally apply weighted sampling, the WLS objective and batch-level MixUp/CutMix. Keep validation and test deterministic.
9: If validation is class-complete, select by validation loss with early stopping; otherwise train for the same pre-specified fixed epoch budget used by every compared configuration.
10: Evaluate once on the untouched test subset; store predictions, confusion matrix, macro metrics, split hash, configuration signature and checkpoint.
11: **end for**
12: Aggregate the three seeds as mean, standard deviation and Student-t 95% confidence interval. Never select the best seed.

Remark 1 (Class-completeness and the fixed epoch budget). Step 9 defines class-completeness by the minimum independent-group support used to build the partition in Step 4, not by a non-zero sample count in the realised validation split. The distinction matters for the paired comparison of Section 6.1. The random-split arm's validation subset does contain individual Blast samples, but those samples still derive from only two independent spatial groups, so that arm is not class-complete under this definition either, and both arms consequently train under the fixed epoch budget. The rule was fixed by the protocol before the random split was constructed rather than chosen after its validation composition was observed.

5 Evaluation Protocol

The evaluation protocol is stated separately from the method because several of the conclusions below follow from protocol decisions rather than from modelling decisions. Every decision in this section was fixed before the test subset was read.

5.1 Seeds, budget and model selection

Every configuration is trained with three fixed seeds (42, 123, 2024) and reported as mean $\pm$ standard deviation with a Student-t 95% confidence interval. No best seed is reported anywhere in this study.

Because Blast occupies only two independent groups and is therefore absent from validation (Section 3.3), validation-loss-based checkpoint selection would be scored on a three-class subset. Rather than allow an incomplete criterion to influence some configurations more than others, a fixed budget of 60 epochs is applied to every compared configuration. The budget was set from the convergence behaviour of pilot training and validation curves, never from test metrics. The constraint applies identically to all architectures and both strategies, so it cannot favour any of them, and the same budget is used in both partition arms, which is what makes the leakage comparison of Section 6.1 interpretable.

5.2 Learning-rate fairness audit

Comparing architectures under a single shared learning rate conflates architectural merit with compatibility with that particular rate. The risk is not hypothetical. In an earlier configuration of this study a rate of 4×10^{-6}, appropriate for the LayerNorm-based transformers, was applied to EfficientNet-B3, whose BatchNorm statistics are additionally destabilised by the small batch size that this input resolution requires. The resulting model reached only 0.213 macro-F1. This is not described here as "below chance": macro-F1 has no single chance level under class imbalance, and the expected macro-F1 of a uniform-random classifier on this test distribution (65/8/98/9 across four classes) is approximately 0.199 rather than 0.25, because per-class precision under random guessing tracks class prevalence. The figure of 0.213 is better read from the training curves, which show train loss failing to leave the region near $\ln 4 \approx 1.386$ for many epochs. It indicates severe under-training and a near-degenerate baseline rather than a property of the architecture, and reporting it as a baseline would have misrepresented the comparison.

A pre-specified search was therefore conducted under rules fixed in advance. The grid $\{4 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$ is applied identically to all six architecture×strategy cells, so no architecture receives a customised grid. Each candidate is trained for a fixed 20-epoch pilot budget with a single fixed seed (42), giving 24 pilot runs. Selection uses validation loss alone, with validation macro-F1 over the classes present in validation as a tie-break. The pilot trainer is implemented as a separate routine that constructs no test loader, so the test subset is unreachable from the search by construction rather than by convention. The complete search table, including every rejected candidate, is reported in Table 4. Only cells whose selected rate

Table 4 Complete learning-rate search. Every candidate is trained for the same 20-epoch pilot budget with seed 42 and scored on the validation subset only; the test subset is unreachable from this procedure by construction. Selected values in bold. Rejected candidates are shown deliberately, as evidence against selective reporting. No candidate diverged.

Configuration	Metric	4×10^{-6}	1×10^{-5}	5×10^{-5}	1×10^{-4}
ViT-B/16 + CE	Val. loss	0.170	**0.142**	0.219	0.477
	Val. $F1_M$	0.944	**0.987**	0.979	0.804
ViT-B/16 + WLS+MSDA	Val. loss	0.354	0.333	**0.318**	0.437
	Val. $F1_M$	0.732	0.799	**0.890**	0.570
Swin-T + CE	Val. loss	0.243	0.201	**0.149**	0.221
	Val. $F1_M$	0.931	0.926	**0.950**	0.929
Swin-T + WLS+MSDA	Val. loss	0.454	0.413	**0.322**	0.327
	Val. $F1_M$	0.520	0.627	**0.828**	0.851
EfficientNet-B3 + CE	Val. loss	1.051	0.669	0.340	**0.264**
	Val. $F1_M$	0.580	0.683	0.901	**0.923**
EfficientNet-B3 + WLS+MSDA	Val. loss	1.029	0.954	0.519	**0.434**
	Val. $F1_M$	0.288	0.355	0.496	**0.529**

differs from the previously used value are retrained under the full three-seed, 60-epoch protocol; unchanged cells reuse the earlier result after signature checking.

The validation figures in Table 4 are not comparable with the test figures reported in Section 6. Blast is absent from the validation subset, so validation macro-F1 is computed over three classes on a subset from which the hardest class is missing. A validation macro-F1 above 0.9 therefore coexists with a test macro-F1 near 0.5 for the same configuration, and the gap reflects the difference between the two label spaces rather than overfitting. This is why validation loss, not validation macro-F1, is the selection criterion, and why no validation figure is interpreted anywhere as an estimate of held-out performance.

No candidate diverged. The selected rate changed in five of the six cells (Table 4), which is itself the substantive result, since it establishes that a shared learning rate was not neutral with respect to the comparison. Two patterns are visible. The imbalance-oriented strategy prefers a higher rate than plain cross-entropy on the same backbone, 5×10^{-5} against 1×10^{-5} for ViT-B/16, so holding the rate fixed while switching the strategy had been confounding strategy with optimiser setting. The BatchNorm-based EfficientNet-B3 selects the highest rate on the grid under both strategies, consistent with the diagnosis above.

Remark 2 (Disclosed boundary condition). Both EfficientNet-B3 cells select the largest rate on the grid, so their optimum may lie marginally beyond 1×10^{-4}. The grid was not extended, because doing so after observing the outcome would break the pre-specification that gives the search its evidential value. The reported EfficientNet-B3 figures should therefore be read as a defensible lower bound rather than a tuned optimum.

5.3 Metrics and uncertainty

Given the imbalance, overall accuracy is reported together with macro-averaged F1, which weights all four classes equally and is therefore the primary criterion. Confidence intervals use the Student-t distribution with $n = 3$. Such an interval is wide by

construction at three seeds, and overlapping intervals are interpreted throughout as an absence of evidence for a difference rather than as evidence of equivalence. The leakage comparison of Section 6.1 is additionally reported as a paired difference, matched by configuration and seed, so that random initialisation is held constant within each pair and the paired t-test operates on the difference distribution directly.

5.4 Efficiency measurement

Efficiency is measured on an NVIDIA A100-SXM4-40 GB under CUDA automatic mixed precision, with batch size 1 and input $1 \times C \times 224 \times 224$, using 20 warm-up iterations followed by 100 timed iterations with explicit device synchronisation before and after each measurement, and with peak-memory counters reset immediately before timing. One inference unit is one Sentinel-2 patch of 264×264 pixels resized to 224×224, corresponding to approximately 2.8 km $\times$ 2.8 km.

The reported latencies cover model inference only; band reading, resampling and normalisation are excluded, since they are common to all architectures and depend on storage rather than on the model.

6 Results

6.1 How much does image-level random splitting inflate performance?

Three configurations (ViT-B/16 with CE, ViT-B/16 with WLS+MSDA, and Swin-T with CE) are trained on both partitions with everything else held identical, and the difference is taken pairwise within each seed.

The effect is large and consistent in sign across all nine seed-level pairs (Table 5, Figure 2). Random image-level splitting inflates macro-F1 by +0.273 for ViT-B/16 with CE, +0.155 for Swin-T with CE and +0.138 for ViT-B/16 with WLS+MSDA, which corresponds to relative overstatement of 52.6%, 25.9% and 25.3% respectively. All three 95% confidence intervals exclude zero, and all three nominal paired t-tests, one per configuration and unadjusted, reject at $p < 0.05$. Treating the three tests as a family and applying a Holm correction, one configuration (ViT-B/16 + WLS+MSDA, nominal $p = 0.013$) remains significant after Holm correction (adjusted $p = 0.038$), while the other two are borderline (adjusted $p = 0.053$). Both values appear in Table 5.

The consistent direction and magnitude across all nine observed seed-level pairs, rather than any single significance threshold, is the more informative descriptive evidence for the effect, although these pairs are not fully independent observations, since they share the same cohort, seeds and related configurations.

Table 5 Leakage effect, paired by configuration and seed. Everything except the partition rule is identical between the two arms: cohort, code, seeds, epoch budget, learning rate, augmentation, and per-arm train-only normalisation. Positive inflation means the conventional random protocol reports a higher score than the leakage-controlled protocol on the same data and the same model. Nominal p is unadjusted (one test per configuration); Holm p treats the three tests as a family.

Configuration	Grouped $F1_M$	Random $F1_M$	Inflation (paired)	95% CI	Nominal p	Holm p
ViT-B/16 + CE	0.518	0.791	+0.273	[0.078, 0.468]	0.027	0.053
Swin-T + CE	0.600	0.755	+0.155	[0.011, 0.300]	0.044	0.053
ViT-B/16 + WLS+MSDA	0.544	0.682	+0.138	[0.071, 0.205]	0.013	0.038

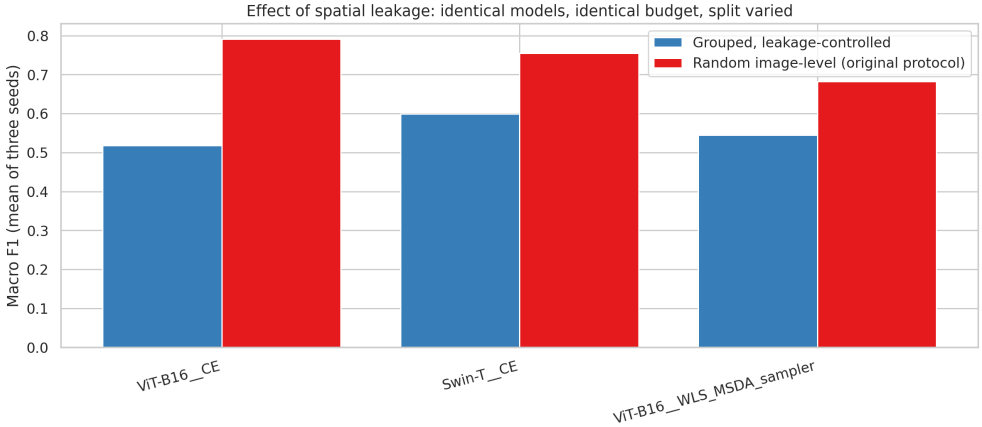


Fig. 2 Macro-F1 under the leakage-controlled and the conventional random partitions, for three configurations across three seeds.

Two aspects of the result deserve emphasis. The sign is the same in all nine seed-level pairs, so the effect is not an artefact of a favourable seed. The magnitude also varies systematically with the training strategy: the configuration with the strongest regularisation (WLS+MSDA) shows the smallest inflation, and the least regularised configuration the largest. A plausible reading is that heavy regularisation limits the extent to which a model can exploit near-duplicate training patches, leaving less leaked signal available to convert into apparent test performance. This is offered as an interpretation consistent with the data rather than as an established mechanism.

The practical implication is direct. An earlier configuration of this study, evaluated on a single seed under the random protocol, reported 0.817 macro-F1, and the best random-protocol result under the present three-seed protocol is 0.791. Under leakage control on identical data, the same study reaches 0.600. Any comparison of published Sentinel-2 disease-classification results that mixes the two protocols is therefore comparing quantities that differ by more than the margins usually reported between competing methods.

Remark 3 (Scope). The magnitude reported here is specific to this benchmark, whose 900 patches occupy only 97 independent spatial groups. It should be read as *leakage-induced inflation on this benchmark*, not as a transferable constant. What generalises is the requirement to measure the effect, not the number.

6.2 Architecture and strategy under matched learning rates

Table 6 reports the full factorial comparison on the group-disjoint partition after the learning-rate audit, and Figure 3 shows the corresponding intervals. Classical learners on handcrafted spectral descriptors, namely support vector machines [53], random forests [54] and gradient-boosted trees [55, 56], were evaluated during preliminary work as a check on whether per-band statistics, vegetation indices and band ratios suffice for this task. They do not; every such configuration failed on the two rare classes. They are therefore treated as handcrafted-feature diagnostics, and only learned representations appear in the main comparison.

Table 6 Architecture × strategy on the leakage-controlled partition (Protocol A, three seeds, 60 epochs, learning rate selected per cell on validation data only). Values are mean ± SD. Student-t 95% confidence intervals are shown in Figure 3. The final column gives the change relative to the same cell trained at the previously shared learning rate.

Backbone	Strategy	LR	Accuracy	$F1_M$	$\Delta F1_M$
Swin-T	CE	5×10^{-5}	0.876 ± 0.021	$\mathbf{0.600 \pm 0.088}$	+0.009
Swin-T	WLS+MSDA	5×10^{-5}	0.806 ± 0.042	0.578 ± 0.034	+0.074
ViT-B/16	WLS+MSDA	5×10^{-5}	0.756 ± 0.020	0.544 ± 0.017	+0.019
EfficientNet-B3	CE	1×10^{-4}	0.837 ± 0.027	0.533 ± 0.011	+0.041
ViT-B/16	CE	1×10^{-5}	0.848 ± 0.008	0.518 ± 0.069	±0.000
EfficientNet-B3	WLS+MSDA	1×10^{-4}	0.596 ± 0.036	0.472 ± 0.035	+0.259

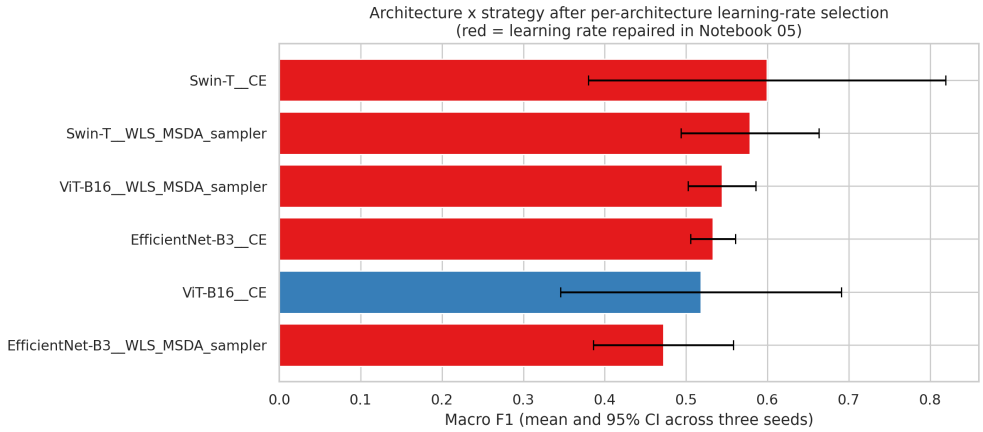


Fig. 3 Macro-F1 with Student-t 95% confidence intervals for the six architecture×strategy cells on the leakage-controlled partition.

Three observations follow, each stated at the strength the data supports.

The learning-rate audit changes the comparison materially.

Every one of the five retrained cells improves. The largest change is EfficientNet-B3 with WLS+MSDA, which moves from 0.213, a severely under-trained and near-degenerate baseline, to 0.472, an increase of +0.259 macro-F1 obtained purely by correcting the optimiser setting. Swin-T with WLS+MSDA gains +0.074. Had the shared rate been retained, this study would have reported a CNN baseline that was not trained and a Swin configuration that was undertrained, and both errors would have flattered the transformer configurations originally proposed.

No architecture is separated from the others at three seeds.

Swin-T with CE attains the highest mean macro-F1 (0.600), but its 95% confidence interval, [0.380, 0.819], overlaps every other cell in the table, including the lowest. No architectural winner is therefore claimed. The two widest intervals belong to the two CE cells, whose per-seed variability is driven by the small test support of the rare classes (Rust $n = 9$, Blast $n = 8$) rather than by unstable optimisation; validation losses for all three seeds of Swin-T + CE converge to the range 0.13–0.18.

The imbalance-oriented strategy does not dominate.

With matched learning rates it helps on one backbone and hurts on the other two: on ViT-B/16 it raises macro-F1 from 0.518 to 0.544, on Swin-T it lowers it from 0.600 to 0.578, and on EfficientNet-B3 it lowers it from 0.533 to 0.472. The direction of the effect is therefore inconsistent across backbones, and every one of these differences lies inside overlapping confidence intervals. This corrects the earlier claim that the training strategy is the dominant performance factor; once the learning rate is matched and leakage is removed, that claim is no longer supported.

6.3 Per-class behaviour

The aggregate metrics in Table 6 conceal a structural failure that the per-class scores make visible. Swin-T with cross-entropy, the configuration with the highest mean macro-F1, recovers at most one of the eight Blast test samples in any seed: its Blast recall is 0.125, 0.000 and 0.000 across the three seeds, a mean of 0.042, and the Blast samples it misses are absorbed almost entirely by RPH (Figure 5, Table 7). Its macro-F1 of 0.600 is therefore composed of competent performance on three classes together with near-total failure on the fourth, whose per-class F1 is 0.061. The same pattern affects every configuration in the table: mean Blast recall ranges from 0.00 to 0.08 under cross-entropy and reaches 0.46 only for EfficientNet-B3 with the imbalance-oriented strategy, whose macro-F1 is the lowest of the six.

Table 7 Per-class F1 on the leakage-controlled test subset, mean ± SD over three seeds. Test support: Aphid 65, Blast 8, RPH 98, Rust 9.

Configuration	Aphid	Blast	RPH	Rust
Swin-T + CE	0.905 ± 0.015	0.061 ± 0.105	0.937 ± 0.007	0.496 ± 0.243
Swin-T + WLS+MSDA	0.861 ± 0.023	0.067 ± 0.115	0.895 ± 0.029	0.491 ± 0.070
ViT-B/16 + WLS+MSDA	0.826 ± 0.013	0.126 ± 0.067	0.866 ± 0.029	0.359 ± 0.072
EfficientNet-B3 + CE	0.866 ± 0.014	0.000 ± 0.000	0.912 ± 0.026	0.354 ± 0.021
ViT-B/16 + CE	0.860 ± 0.017	0.112 ± 0.098	0.932 ± 0.006	0.169 ± 0.190
EfficientNet-B3 + WLS+MSDA	0.725 ± 0.115	0.203 ± 0.055	0.724 ± 0.026	0.237 ± 0.049

This is a consequence of the benchmark rather than of any model. Blast occupies two independent spatial groups (Section 3.3), so under group disjointness its training support consists of near-replicate patches drawn from two locations, and its test support is eight samples from neither. A model has almost no basis on which to learn a location-independent Blast signature. The random-split protocol concealed this by supplying test patches drawn from the same groups as the training patches, which is the class-specific form of the effect quantified in Section 6.1.

Accuracy and macro-F1 order the configurations differently, and the divergence is informative rather than anomalous. ViT-B/16 with cross-entropy attains the second-highest accuracy in Table 6 (0.848) alongside the second-lowest macro-F1 (0.518). Aphid and RPH together account for 163 of the 180 test samples, so a model that classifies those two classes well and neglects Blast and Rust can still reach an accuracy near 0.85. Macro-F1 removes that shelter by weighting all four classes equally, which is why it is the primary criterion here. Accuracy is retained in the tables because its divergence from macro-F1 is itself diagnostic: where the two disagree sharply, the model is trading rare-class recall for majority-class precision.

Two consequences follow for how Table 6 should be read. First, "highest mean macro-F1" identifies the configuration that degrades least under this constraint, not one that solves the four-class problem; the term is used in that restricted sense throughout. Second, the ordering of the two training strategies depends on which failure is weighted more heavily. Cross-entropy achieves higher aggregate scores while abandoning the rarest class; the imbalance-oriented strategy recovers some Blast recall at a

cost in overall accuracy. Neither behaviour is preferable in the abstract, and neither is presented as such here.

6.4 Normalisation and patch-embedding treatment

Two design choices carried over from prior practice, ImageNet-replicated normalisation and a frozen patch embedding, were previously justified by argument rather than by measurement. Table 8 reports the 2×2 ablation that tests them, with all other settings fixed and three seeds per cell.

Table 8 Normalisation × patch-embedding ablation (ViT-B/16, Protocol A, leakage-controlled partition, three seeds). All four cells use WLS+MSDA with weighted sampling and a learning rate of 4×10^{-6}; normalisation and patch-embedding treatment are the only factors varied. Per-band statistics are computed from the training subset only.

Normalisation	Patch embedding	Accuracy	$F1_M$	$F1_M$ 95% CI
Per-band (train-only)	Frozen	0.746 ± 0.032	$\mathbf{0.546 \pm 0.027}$	[0.480, 0.612]
Per-band (train-only)	Fine-tuned	0.733 ± 0.015	0.531 ± 0.032	[0.451, 0.611]
ImageNet-replicated	Fine-tuned	0.698 ± 0.018	0.520 ± 0.010	[0.495, 0.545]
ImageNet-replicated	Frozen	0.674 ± 0.006	0.499 ± 0.013	[0.467, 0.530]

Per-band normalisation gives consistently higher mean macro-F1. It exceeds ImageNet-replicated statistics under both embedding treatments, 0.546 against 0.499 with a frozen embedding and 0.531 against 0.520 when fine-tuned, although the fine-tuned margin is small and has not been tested with a paired significance test. Accuracy shows the same ordering with a larger margin, 0.746 against 0.674 when frozen. The result is physically unsurprising, since NIR reflectance on this cohort averages 0.378 against 0.235 for blue, so a single RGB-derived mean and standard deviation misplaces the NIR and SWIR channels. It had not previously been demonstrated, however, and per-band normalisation is adopted here on evidence rather than on argument.

Freezing versus fine-tuning the patch embedding is not resolved. With per-band statistics the frozen variant is ahead (0.546 against 0.531); with ImageNet statistics the fine-tuned variant is ahead (0.520 against 0.499). Both differences lie within approximately one standard deviation at $n = 3$, so this is reported as a descriptive trend and no significant effect is claimed. Figure 4 supports the same reading visually: B8-isolated patch-embedding activations for matched samples are near-identical before and after fine-tuning, which indicates that with only 641 training patches the layer barely moves from its initialisation, and explains why the choice matters little.

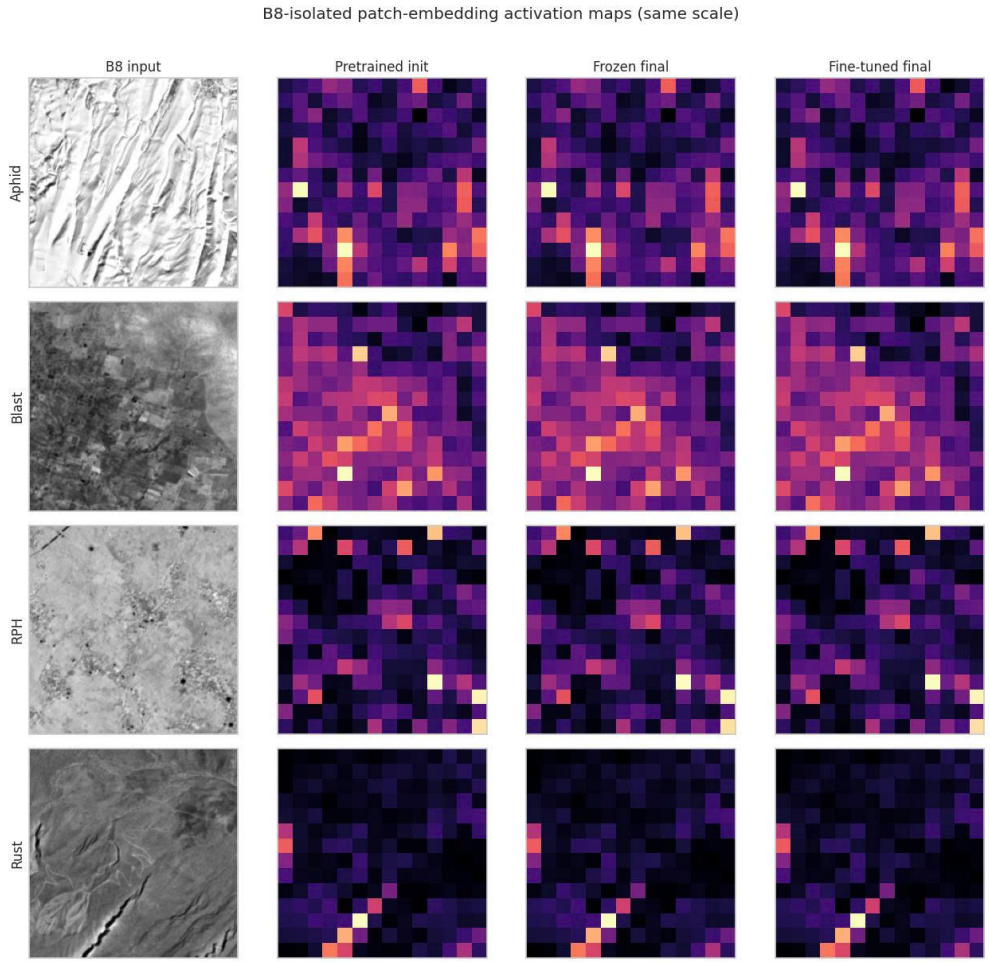


Fig. 4 B8-isolated patch-embedding activation maps for one representative sample per class, on a common colour scale. Columns: B8 input; activations at the replicated pretrained initialisation; after training with the embedding frozen; after training with it fine-tuned.

6.5 Contribution of mixed-sample augmentation

Earlier work associated mixed-sample augmentation with a large improvement. Table 9 isolates its contribution within the WLS branch, holding the backbone, loss, sampler and learning rate constant across the two rows, at the rate Section 5.2 selected for this branch (5×10^{-5}).

The difference in macro-F1 is +0.016 in favour of MSDA, well inside one standard deviation on both sides (SD = 0.068 without MSDA, 0.017 with it), while accuracy and weighted F1 remain slightly higher without it. Mixed-sample augmentation therefore provides no measurable macro-F1 benefit on this benchmark under leakage control.

Table 9 Effect of mixed-sample augmentation within the WLS branch (ViT-B/16, Protocol A, leakage-controlled partition, three seeds, identical learning rate 5×10^{-5} in both rows).

Configuration	Accuracy	$F1_M$	$F1_W$
WLS, no MSDA	$\mathbf{0.806 \pm 0.024}$	0.528 ± 0.068	$\mathbf{0.809 \pm 0.023}$
WLS + MSDA	0.756 ± 0.020	0.544 ± 0.017	0.793 ± 0.014

This substantially revises the earlier reading, and the explanation is instructive: the large apparent gain previously attributed to MSDA was measured on a random partition, where, as Section 6.1 establishes, a substantial part of any measured difference is attributable to the partition rather than to the treatment.

An earlier version of this comparison used 4×10^{-6} in both rows, the shared rate later found suboptimal for the WLS strategy, and reported a difference of +0.006. That comparison was internally fair, since the rate was identical in both rows, but both configurations were rerun at the selected rate so that the conclusion no longer rests on that argument. The conclusion is unchanged.

6.6 Band parsimony: four bands versus six

Protocol B removes the red-edge (B5) and SWIR (B11) channels while holding the cohort, the partition, the model, the strategy and the three seeds constant, so the comparison isolates the contribution of those two bands (Table 10). This comparison predates the learning-rate audit of Section 5.2 and was not rerun once the audit selected 5×10^{-5} for this configuration elsewhere. Both rows instead use the shared pre-audit rate 4×10^{-6}, applied identically to both band counts, so that the four-versus-six-band difference is not itself confounded by rate. The six-band row below therefore differs from the 0.544 macro-F1 reported for the nominally identical configuration in Table 6 and Table 9 by learning rate alone, not by any change in data, partition or seeds.

Table 10 Six-band versus four-band configuration (ViT-B/16 + WLS+MSDA, identical 900-patch cohort, identical group-disjoint partition, three seeds, shared pre-audit learning rate 4×10^{-6} in both rows; see text).

Configuration	Accuracy	$F1_M$	$F1_M$ 95% CI	$F1_W$
Six bands (B2,B3,B4,B5,B8,B11)	0.746 ± 0.027	0.525 ± 0.033	[0.443, 0.607]	0.784
Four bands (B2,B3,B4,B8)	0.750 ± 0.025	0.510 ± 0.032	[0.429, 0.590]	0.786

The four-band configuration matches the six-band configuration to within 0.015 macro-F1, less than one standard deviation and with almost completely overlapping confidence intervals, and the two are indistinguishable on accuracy and weighted F1. Adding red-edge and SWIR channels therefore yields no measurable benefit on this benchmark.

The result is protected against post-hoc selection, since the four bands were fixed before training by band availability and a pre-specified agronomic ordering rather than chosen because they performed well. Its practical value lies in acquisition and storage

rather than computation: as Section 6.7 shows, the two configurations differ by only 0.5% in parameters and FLOPs, since only the stem changes.

Remark 4 (Scope). This finding applies to this benchmark, at 10 m ground sampling distance, with 900 patches and four disease classes. It should not be read as evidence that red-edge or SWIR reflectance is uninformative for plant disease in general; a cohort of this size may simply be too small for the model to exploit the additional channels.

6.7 Computational efficiency

Table 11 reports the efficiency measurements requested for deployment assessment.

Table 11 Computational efficiency on an NVIDIA A100-SXM4-40 GB under CUDA AMP with batch size 1. Latency is measured over 100 timed iterations after 20 warm-up iterations with device synchronisation. One inference unit is a single 224×224 patch ($\approx 2.8\,\text{km} \times 2.8\,\text{km}$); preprocessing is excluded. All parameters are trainable in the primary runs; the frozen patch embedding is used only in the ablation of Table 8.

Protocol	Model	Params (M)	FLOPs (G)	Latency (ms)	Throughput (patch/s)	Peak mem. (MB)
A (6 bands)	ViT-B/16	86.39	16.98	6.60 ± 0.27	151.6	533
	Swin-T	27.53	4.52	14.06 ± 0.72	71.1	195
	EfficientNet-B3	10.70	1.00	16.01 ± 0.98	62.5	91
B (4 bands)	ViT-B/16	86.00	16.91	6.80 ± 0.30	147.1	532
	Swin-T	27.52	4.51	13.91 ± 0.58	71.9	195
	EfficientNet-B3	10.70	0.99	17.34 ± 1.48	57.7	90

Two points are worth drawing out. FLOPs do not predict latency on this hardware: ViT-B/16 requires 3.8× the FLOPs of Swin-T and 17.0× those of EfficientNet-B3, yet is the fastest of the three by more than a factor of two. Dense matrix multiplication maps efficiently onto tensor cores, whereas windowed attention and the sequential depthwise and squeeze-excitation chains of EfficientNet issue many small kernels. Practitioners selecting a model for operational monitoring should therefore benchmark on target hardware rather than rank by FLOPs.

Moving from six bands to four reduces parameters and FLOPs by only about 0.5%, because only the stem projection changes. The benefit of the four-band configuration lies in data acquisition, transfer and storage, a one-third reduction in channels retrieved and archived, rather than in inference cost.

Remark 5 (Measurement caveat). FLOPs are counted with `fvcore`, which has no registered counter for scaled dot-product attention, GELU, softmax, SiLU and elementwise operations. Reported values are therefore a conservative lower bound; the omission is small for these architectures at sequence length 197 but is disclosed for completeness.

6.8 Error structure, posterior probability and calibration

Figure 5 shows aggregate normalised confusion matrices over the three seeds. The dominant error mode is confusion of minority classes with RPH, the majority class, which is the expected consequence of a 12.4:1 imbalance combined with group-disjoint evaluation that removes the near-duplicate support the rare classes previously received.

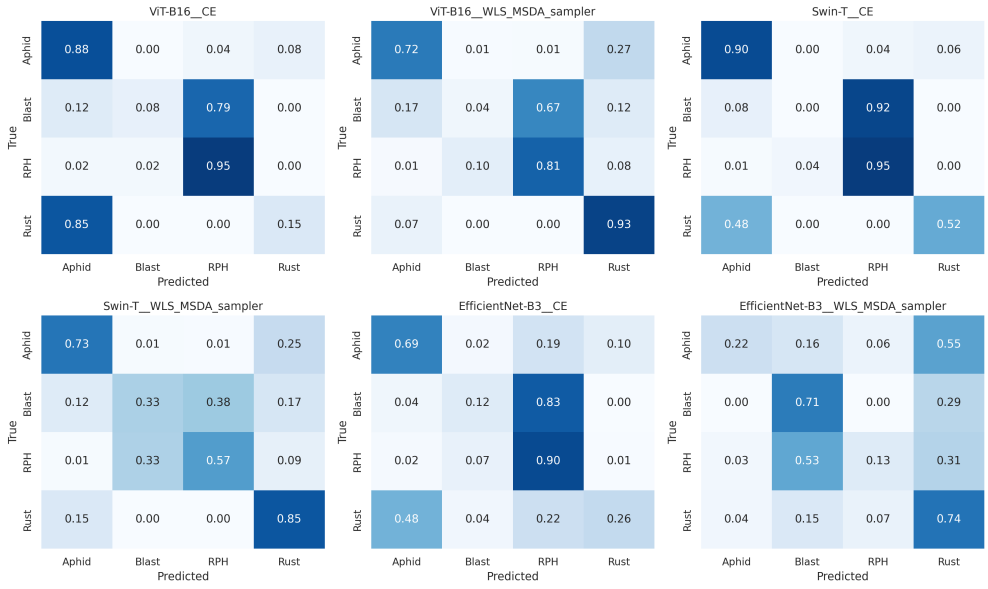


Fig. 5 Aggregate row-normalised confusion matrices on the leakage-controlled test subset, averaged over three seeds.

A hard-label confusion matrix discards the confidence with which errors are made, so two further views are reported. The class-wise mean posterior probability (Figure 6) gives, in each cell, the mean predicted probability assigned to a candidate class over held-out samples of a given true class. The spatial probability-sensitivity map (Figure 7) replaces each 16×16 patch with the training-set mean and records the resulting drop in predicted class probability. The latter is an occlusion-style attribution in the tradition of perturbation-based explanation methods [57], of which LIME [58] and SHAP [59] are the best-known instances. The direct occlusion form is used here because it requires no surrogate model, and because its units, a change in predicted probability, are interpretable without further assumptions.

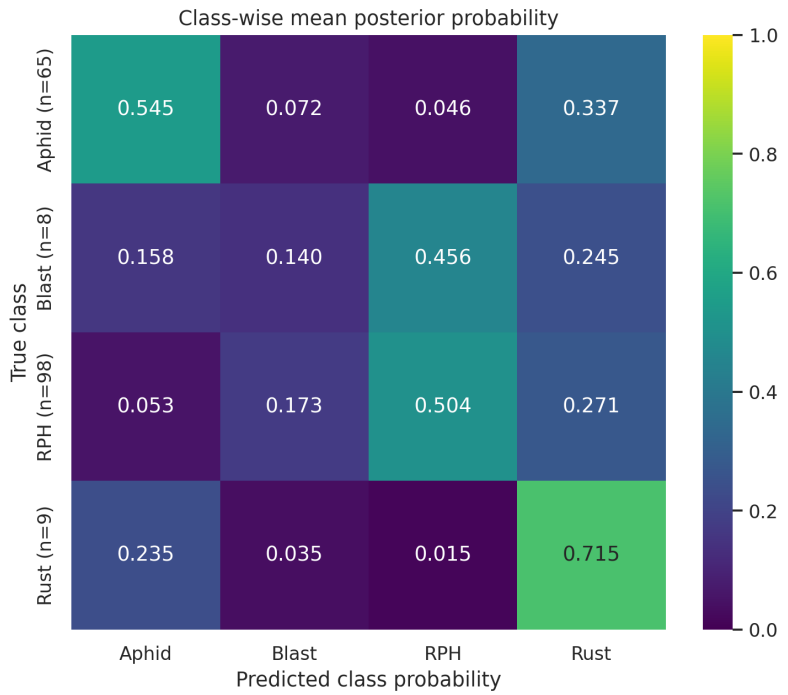


Fig. 6 Class-wise mean posterior probability on the held-out subset. Rows are the true class, columns the predicted class.

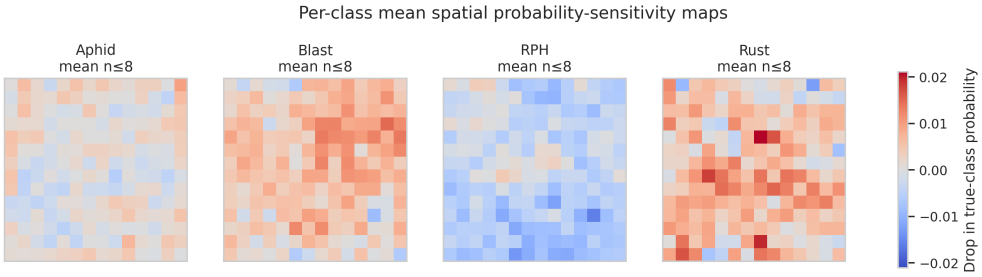


Fig. 7 Class-wise mean spatial probability sensitivity. Brighter regions indicate image areas on which the prediction depends most strongly.

Calibration is reported because a probability used for operational triage must be trustworthy, not merely rank-correct. The expected calibration error is 0.209 under Protocol A and 0.192 under Protocol B (Figure 8). These values are high: stated confidence departs from empirical accuracy by roughly 20 percentage points on average. Raw posteriors should therefore not be used as decision thresholds without recalibration [60]. Post-hoc temperature scaling is the natural next step, but it was not pursued

here because it requires a class-complete validation subset, which this partition cannot provide.

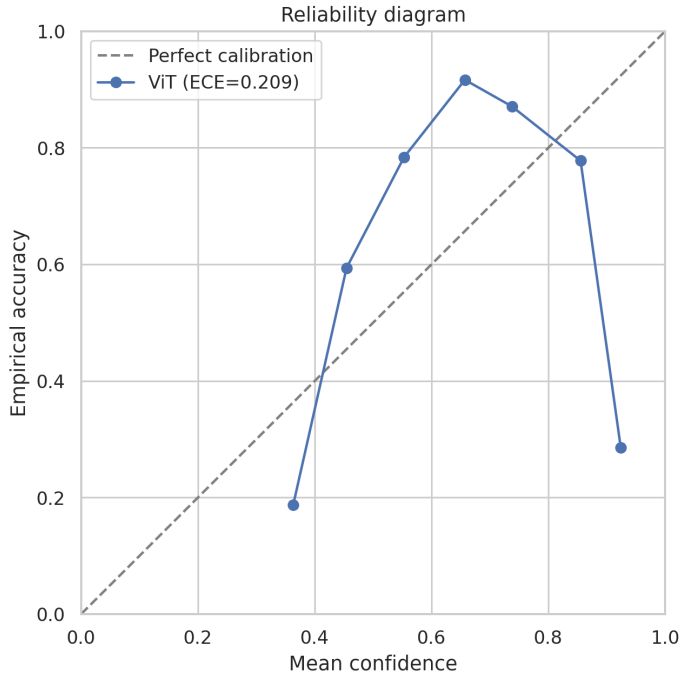


Fig. 8 Reliability diagram on the leakage-controlled test subset (Protocol A).

6.9 Representation quality

To assess whether the network learns a representation better structured than the raw spectral input, t-SNE embeddings [61] of per-band means are compared against those of the learned class-token features, with the comparison quantified by the silhouette coefficient [62] computed in the original standardised feature space rather than in the two-dimensional projection (Figure 9).

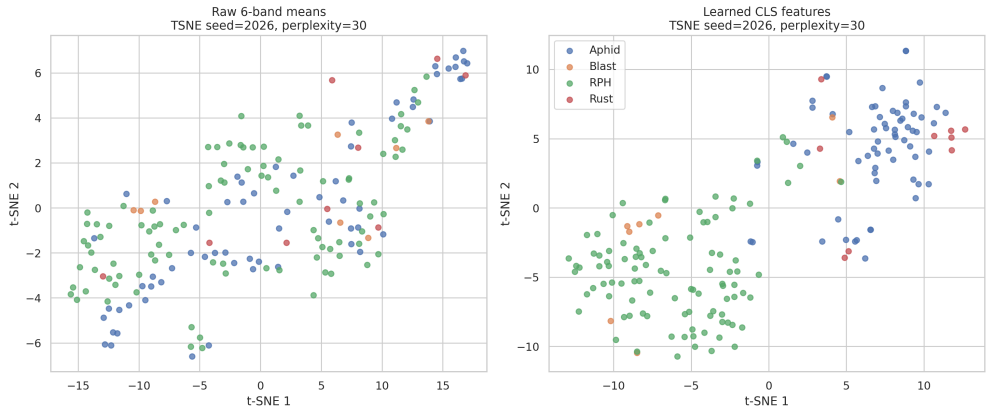


Fig. 9 t-SNE projections of raw six-band per-sample means (left) and learned class-token features (right) on the held-out subset.

The silhouette coefficient moves from -0.071 for raw band means to $+0.066$ for learned features under Protocol A, and from -0.050 to $+0.028$ under Protocol B. The negative values for raw features indicate that classes are not separable in raw spectral space at all, since samples are on average closer to another class's centroid than to their own; the network therefore does learn a better-structured representation. The positive values remain small, so the improvement is characterised here as real but modest. The projection itself is not presented as evidence of separability: t-SNE geometry is a visualisation, and the claim rests on the coefficient computed before projection.

7 Discussion

7.1 What changes when leakage is controlled

The measurement in Section 6.1 has a consequence beyond this dataset. On a benchmark whose 900 patches occupy only 97 independent spatial groups, the conventional evaluation protocol overstates macro-F1 by 25–53%. That exceeds the margins by which competing methods are typically separated in this literature, so a ranking established under random splitting can be dominated by how much each method benefits from leakage rather than by how well it detects disease. The results reported here illustrate the point directly. The configuration that gains most from the random protocol, ViT-B/16 with cross-entropy at $+0.273$, is the one that gains least from it under grouped evaluation, where it ranks last among the three tested cells.

This is consistent with the wider literature on spatially structured validation. Ploton et al. [10] report that ecological mapping models with strong random-split statistics lose most of their apparent skill under spatial validation, Karasiak et al. [11] demonstrate the same bias for remote-sensing classification, and Kapoor and Narayanan [12] identify train–test dependence as a recurrent cause of irreproducible findings across

scientific fields. The magnitude is also consistent with the closest deep-learning precedent. Kattenborn et al. [15] report up to 28% inflation for convolutional networks under random versus spatial cross-validation, against the 25–53% measured here. That this benchmark sits at the upper end of that range is unsurprising given its clustering, and it suggests that the effect scales with how tightly a benchmark is spatially concentrated rather than being a fixed property of the model class. The contribution of the present study is to convert that general warning into a measured, confidence-bounded quantity for satellite-based crop disease classification, under a design in which the partitioning scheme is the sole intentionally manipulated design factor.

What this does not imply is worth stating as plainly. It does not imply that previously published results were incorrectly computed, nor that random splitting is never informative; it remains a reasonable estimate of in-distribution performance when new patches are drawn from fields already represented in training. It implies that random-split figures should not be described as evidence of transferable disease detection, and that results obtained under the two protocols should not be compared with one another.

7.2 Learning rate as a confounder in architecture comparison

The audit in Section 5.2 reproduces in a remote-sensing setting what Wightman et al. [14] established for ImageNet classification when ResNet training recipes were modernised: much of what appears to be an architectural gap can be a gap in training configuration. Here a single shared learning rate left EfficientNet-B3 with WLS+MSDA severely under-trained, and correcting only that setting recovered +0.259 macro-F1. Had the shared rate been retained, this study would have published a CNN baseline that had not converged, which is the pattern that makes a comparative study unreliable.

The mechanism is architectural rather than incidental. ViT and Swin use LayerNorm, which normalises per sample and is insensitive to batch size. EfficientNet uses BatchNorm, whose running statistics are noisy at the small batch size that $224 \times 224 \times 6$ inputs impose, and weighted sampling makes each batch less class-diverse still. The two families therefore have genuinely different optimiser requirements. Holding the learning rate fixed across them does not make the comparison fairer; it makes it fair in appearance while being unfair in substance. Schratz et al. [19] identify the same interaction between tuning and resampling for spatial data, where selection performed under one regime and reporting under another leaves the comparison with a bias that neither step reveals on its own. Comparative studies in this area should therefore either tune the rate per architecture on validation data drawn from the protocol used for reporting, publishing the complete search, or state explicitly that the reported baselines were not individually tuned.

7.3 Two design choices revised on evidence

Section 6.4 replaces an argument with a measurement. The original rationale for ImageNet-replicated normalisation was that a frozen, ImageNet-pretrained patch embedding expects ImageNet-normalised inputs. The reasoning is coherent but was

never tested, and the test does not support it: per-band statistics computed from the training subset are better under both embedding treatments. The physical explanation is that NIR reflectance averages 0.378 on this cohort against 0.235 for blue, so a single RGB-derived mean and standard deviation systematically misplaces the non-visible channels. The associated question of whether to freeze the embedding remains unresolved at three seeds, and Figure 4 suggests why: with 641 training patches the layer barely departs from its initialisation, so the choice has little to act on.

Section 6.5 revises a second claim. The large improvement previously attributed to mixed-sample augmentation is not reproduced under leakage control. At a learning-rate-matched comparison, the paired difference is +0.016 macro-F1, well inside one standard deviation, with accuracy and weighted F1 marginally higher without augmentation. Taken together with the architecture results in Section 6.2, where the imbalance-oriented strategy helps one backbone and hurts two others, the earlier conclusion that training strategy dominates architecture is not supported by the corrected experiments, and we withdraw it. Rana et al. [47] report architecture-dependent behaviour under strong data transformation, which we take as supporting context rather than as a general rule, since the task and transformation differ from those studied here.

7.4 What the benchmark can and cannot support

The per-class results in Section 6.3 bound the interpretation of everything above. No configuration tested here learns Blast: the class occupies two independent spatial groups, so once near-replicate support is removed, a model has no basis on which to learn a signature that transfers to a third location. The aggregate macro-F1 figures therefore describe performance on a task that is, for most configurations, close to three-class. This is a property of the cohort rather than of any architecture or training strategy, and no choice among the options examined here would repair it; additional acquisitions from further locations would.

The same constraint explains why leave-one-region-out evaluation is unavailable (Section 3.6). Disease class and geography are confounded in this cohort to the point that every coordinate-derived regional holdout contains one or two classes rather than four. A benchmark with this structure can support a controlled comparison between evaluation protocols, which is what the present study conducts, but it cannot support a claim of geographic transfer, and reporting one would require a proxy that measures class absence instead.

This bears on what a second dataset would and would not establish. Cross-dataset validation would strengthen a claim about the magnitude of the leakage effect, but the magnitude is not claimed to generalise; Remark 3 states that it is specific to this cohort, and what generalises is the requirement to measure it. The paired design is internal to a single benchmark by construction, since the two arms must share a cohort for the partition rule to be the only variable that differs. A second benchmark would yield a second measurement rather than a confirmation of this one. That said, whether the four-band result and the absence of an MSDA benefit hold elsewhere is genuinely open, and Section 8 records both as unestablished.

7.5 Practical implications

Three implications follow for practitioners. Four visible-to-NIR bands are sufficient on this benchmark, which reduces acquisition and archival cost by a third at no measurable accuracy cost, though not inference cost, which is dominated by the encoder rather than the stem (Section 6.7). FLOPs are a poor proxy for latency on modern accelerators, since ViT-B/16 has 17× the FLOPs of EfficientNet-B3 and runs 2.4× faster, so deployment decisions should rest on measurements taken on the target device. The posteriors are substantially miscalibrated at ECE ≈ 0.2, so they require recalibration before use as operational thresholds; reporting accuracy alone would conceal this.

None of these is a deployment recommendation. A system that does not detect one of four target diseases is not ready for operational use, whatever its aggregate score, and the results here should be read as characterising a benchmark and an evaluation protocol rather than as qualifying a model for the field.

8 Limitations

Several of the limitations below bound the interpretation of the results above, and are ordered accordingly.

One class is effectively unlearnable under group disjointness.

Blast occupies two independent spatial groups, so no configuration tested here attains usable recall on it once near-replicate support is removed (Section 6.3). The macro-F1 figures reported throughout therefore describe performance on a task that is, for most configurations, close to three-class. Establishing whether Blast is separable at this resolution would require acquisitions from further locations rather than a different model or training strategy.

No geographic or temporal generalisation is demonstrated.

Leave-one-season-out evaluation is impossible because no acquisition date is recoverable for any of the 900 patches, and leave-one-region-out evaluation is impossible because all four coordinate-derived candidate regions fail a pre-specified eligibility test, each held-out region containing only one or two of the four classes (Section 3.6). Disease class and geography are confounded in this cohort. This is reported rather than approximated with a weaker proxy that would appear to demonstrate regional transfer while measuring class absence. The group-disjoint protocol should accordingly be understood as removing local spatial dependence, not as establishing transfer to new production regions or new seasons.

No external dataset validation.

A search for a public Sentinel-2 crop-disease dataset with compatible class definitions and acquisition characteristics did not identify one that would support a scientifically valid direct test. A crop-type dataset was not substituted, since crop-type transfer would not evidence disease-level transfer. Cross-dataset generalisation therefore remains unestablished. Section 7 sets out why a second benchmark would yield a second measurement of the leakage effect rather than a confirmation of this one, but the

four-band result and the absence of an MSDA benefit are genuinely open questions that external data would help settle.

Three seeds give wide intervals.

Student-t intervals at $n = 3$ are wide by construction, and the rare-class test support is small (Blast $n = 8$, Rust $n = 9$). Overlapping intervals in Table 6 should therefore be read as an absence of evidence for architectural differences rather than as evidence of equivalence. A larger seed budget, together with bootstrap intervals over test samples in addition to intervals over seeds, would separate seed variance from sampling variance; this is identified here as a direction rather than a completed analysis.

Architecture coverage is bounded by the experimental budget.

Three backbones under two strategies, each at three seeds and preceded by four pilot runs per cell, account for 24 pilot runs and 18 full runs. Widening the comparison to spectral-native architectures such as SpectralFormer [36], or to domain-pretrained models such as SatMAE [9], was not feasible within that budget while retaining per-cell learning-rate selection and three-seed reporting. A broader comparison conducted at one seed under a shared learning rate would reintroduce precisely the two defects that Sections 6.1 and 5.2 identify, so the narrower comparison was preferred.

No cloud masking or quality filtering.

The imagery is used as distributed by the competition organisers, without scene classification masking or quality-band filtering. Part of the per-band variance in Table 3, where standard deviations approach or exceed the means, reflects atmospheric conditions rather than canopy state. Masking would reduce this variance but also the effective cohort size, which is already small, and it was not applied.

Validation is not class-complete.

Blast cannot appear in train, validation and test simultaneously under group disjointness. Its absence from validation was accepted and compensated with a fixed epoch budget applied identically to every configuration. This is a property of the benchmark rather than of the method, but it does mean that validation-based early stopping and post-hoc temperature scaling are unavailable here.

The learning-rate grid is bounded above for one architecture.

Both EfficientNet-B3 cells selected the largest rate on the pre-specified grid, so their optimum may lie slightly beyond 1×10^{-4}. The grid was not extended after the result was observed, in order to preserve the pre-specified character of the search, and the reported EfficientNet-B3 figures should therefore be read as a defensible lower bound.

The band-parsimony comparison predates the learning-rate audit.

Table 10 reuses the shared pre-audit rate 4×10^{-6} for both band counts rather than the 5×10^{-5} rate subsequently selected for this configuration in Section 5.2. The rate is identical across the two rows, so the four-versus-six-band comparison is not itself

confounded by rate, but its six-band figure is consequently not directly comparable with the 0.544 macro-F1 reported for the nominally identical configuration elsewhere. Rerunning the four-band arm at 5×10^{-5} would remove this caveat.

The grouping thresholds are not sensitivity-tested.

The 500 m proximity threshold and the 10 km block size (Section 3.3) were fixed as conservative, physically motivated defaults before the audit was run, and computed after reprojecting every patch to a common planar CRS (EPSG:3857). How the resulting 97 groups, or the leakage estimate that depends on them, would change under alternative thresholds, for example 5 km or 20 km blocks, has not been measured. The rule and its outcome are reported rather than a sensitivity sweep, and this remains a direction for a future audit of this benchmark.

The leakage magnitude is benchmark-specific.

The inflation reported in Section 6.1 reflects the spatial clustering of this particular cohort, 900 patches in 97 groups. The requirement to measure the effect generalises; the number does not.

No temporal modelling and no domain-specific pretraining.

Each patch is classified independently, so disease progression dynamics are not exploited, and temporal encoders [63] are a natural extension. ImageNet initialisation with channel replication likewise remains an imperfect match to multispectral statistics, and domain pretraining such as SatMAE [9], or spectral-native architectures such as SpectralFormer [36], may remove the need for replication entirely.

9 Conclusion

This paper asked what a spectral-adapted Vision Transformer achieves on Sentinel-2 crop disease classification once the evaluation protocol no longer permits spatially dependent patches to appear on both sides of the train/test boundary.

Answering that required first measuring the effect of the conventional protocol. Under a paired design in which the partitioning scheme is the sole intentionally manipulated design factor, with identical cohort, code, seeds, epoch budget, learning rates, augmentation and per-arm train-only normalisation, random image-level splitting inflates macro-F1 by +0.273, +0.155 and +0.138 for the three tested configurations, a relative overestimation of 25–53%. All confidence intervals exclude zero and all paired tests are nominally significant at $p < 0.05$; Section 6.1 reports the Holm-adjusted values, and the consistent direction across all nine seed-level pairs is the evidence we weight most heavily. The two arms differ marginally in train and validation size, 641/79 against 629/91, which is an unavoidable consequence of group-disjoint partitioning rather than a design choice; the test subsets are identical in size. To our knowledge this is the first controlled measurement of the effect on a satellite crop-disease benchmark.

Under leakage control, and after a pre-specified learning-rate search that changed the selected rate in five of six architecture×strategy cells, the highest mean macro-F1

is 0.600 ± 0.088, and no architecture is separated from the others beyond overlapping intervals at three seeds. The audit also showed that a shared learning rate is not a neutral experimental choice, since it had left a BatchNorm-based CNN baseline severely under-trained at 0.213 macro-F1, a figure that correcting the optimiser setting alone lifted to 0.472.

These aggregate figures require one qualification. No configuration tested here attains usable recall on Blast, the class supported by only two independent spatial groups, so the macro-F1 values describe performance on a task that is, for most configurations, close to three-class. That is a property of the cohort rather than of any model, and it is the reason the study characterises a benchmark and an evaluation protocol rather than qualifying a system for operational use.

Two design choices previously justified by argument are revised on evidence. Train-only per-band normalisation gives consistently higher mean macro-F1 than ImageNet-replicated statistics for the NIR and SWIR channels under both patch-embedding treatments, and mixed-sample augmentation yields no measurable macro-F1 benefit once leakage is controlled. A pre-declared four-band configuration matches the six-band configuration within one standard deviation, with no detectable benefit from adding the red-edge and SWIR channels at this resolution and cohort size. That result has direct implications for acquisition cost, but it should be read as an absence of detected difference rather than as proof of equivalence. Leave-one-region-out and leave-one-season-out evaluation are reported as infeasible from the available metadata rather than approximated with a proxy that would measure class absence instead of geographic transfer.

The split manifests, per-seed results and the complete learning-rate search including rejected candidates are released with the paper. The broader recommendation is procedural. For geospatial benchmarks with strong spatial clustering, the leakage-controlled and random-split protocols answer different questions. A study should state which question it is answering and, where the cohort permits, measure the distance between the two.

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Consent to participate and consent for publication. Not applicable.

Data availability. The dataset is distributed by the organisers of the ICPR 2024 *Beyond Visible Spectrum: AI for Agriculture* competition [48]. The split manifests, group assignments, per-seed results, the complete learning-rate search table including rejected candidates, and the leakage-effect summaries generated in this study are available from the corresponding author on reasonable request.

Code availability. The evaluation pipeline, including the grouping and partitioning procedure of Algorithm 1, is available from the corresponding author on reasonable request.

Author contribution. All authors contributed to the study conception, design, experiments and manuscript preparation, and approved the final manuscript.

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